

Initial Value Problem

$$\begin{cases} x'(t) = f(t, x(t)) \\ x(t_0) = x_0 \end{cases} \quad (\text{IVP}) \quad \longleftrightarrow \quad x(t) = x_0 + \int_{x_0}^t f(s, x(s)) ds$$

Lemma 1.3.1. $G := \{(t, x) : |t - t_0| \leq a, |x - x_0| \leq b\}$.

suppose that $f \in C(G, \mathbb{R}^n)$ and is Lipschitz in x on G .

Then IVP has a **unique solution** which exists for $|t - t_0| \leq \gamma$,

where $\gamma = \min\{a, \frac{b}{M}\}$ with $M = \max_{(t,x) \in G} |f(t, x)|$

Pf. (Uniqueness)

Assume (IVP) has two solutions $x_1(t), x_2(t)$.

$$x_i(t) = x_0 + \int_{t_0}^t f(s, x_i(s)) ds, \quad i=1, 2.$$

$$\begin{aligned} \Rightarrow |x_1 - x_2| &= \left| \int_{t_0}^t f(s, x_1) - f(s, x_2) ds \right| \\ &\stackrel{u(t)}{\leq} \int_{t_0}^t |f(s, x_1) - f(s, x_2)| ds \\ &\leq \int_{t_0}^t L |x_1 - x_2| ds \end{aligned}$$

$$\Rightarrow u(t) \leq L \int_{t_0}^t u(s) ds.$$

Since $u \geq 0$, $u(t_0) = 0$, $u \equiv 0$ by Gronwall's inequality.

(Existence)

Define $\{x_m(t)\}_{m=0}^\infty$ by

$$x_{m+1}(t) = x_0 + \int_{t_0}^t f(s, x_m(s)) ds \quad m=0, 1, 2, \dots$$

Step 1. We claim that $x_m(t)$ is defined on the $J = [t_0 - \gamma, t_0 + \gamma]$ and $|x_m(t) - x_0| \leq b$. (i.e. $(t, x_m(t)) \in G$ for $t \in J$).

pf. claim holds for $m=0$. $m \Rightarrow m+1$:

$$(t, x_m(t)) \in G \Rightarrow |f(t, x_m)| \leq M, \quad \forall t \in J.$$

$$\begin{aligned} \Rightarrow |x_{m+1}(t) - x_0| &= \left| \int_{t_0}^t f(s, x_m(s)) ds \right| \\ &\leq \int_{t_0}^t |f(s, x_m(s))| ds \\ &\leq M(t - t_0) \\ &\leq M\gamma \leq b. \end{aligned}$$

Step 2. $\{x_m(t)\}_{m=0}^\infty$ is uniformly convergent on J :

$$x_m(t) = x_0 + \sum_{i=0}^{m-1} (x_{i+1}(t) - x_i(t)).$$

$$\text{Claim: } |x_{i+1} - x_i| \leq \frac{ML^i |t - t_0|^{i+1}}{(i+1)!} \leq \frac{ML^i \gamma^{i+1}}{(i+1)!}$$

holds for $i=0$. $i \Rightarrow i+1$:

$$|x_{i+2} - x_{i+1}| = \left| \int_{t_0}^t f(s, x_{i+1}) - f(s, x_i) ds \right|$$

$$\begin{aligned}
&\leq \int_{t_0}^t L |x_{\bar{i}+1} - x_{\bar{i}}| ds \\
&\leq \int_{t_0}^t L \cdot \frac{ML^{\bar{i}} |s-t_0|^{\bar{i}+1}}{(\bar{i}+1)!} ds \quad (t \geq t_0) \\
&= \frac{ML^{\bar{i}+1} (t-t_0)^{\bar{i}+2}}{(\bar{i}+2)!} \sim \dots
\end{aligned}$$

By Weierstrass M-series criterion,

$$\sum_{\bar{i}=0}^{\infty} \frac{ML^{\bar{i}} \gamma^{\bar{i}+1}}{(\bar{i}+1)!} \text{ is convergent}$$

$$\Rightarrow \sum_{\bar{i}=0}^{\infty} (x_{\bar{i}+1} - x_{\bar{i}}) \text{ is uniformly convergent on } J$$

$$\Rightarrow \{x_m(t)\}_{m=0}^{\infty} \text{ is uniformly convergent on } J$$

$$\Rightarrow x_m \rightarrow x, \quad x(t) \text{ is continuous on } J.$$

Lipschitz $\Rightarrow f(t, x_m)$ is uniformly convergent to $f(t, x)$ on J
 $\Rightarrow x = x_0 + \int_{t_0}^t f(s, x(s)) ds.$

Thm 1.3.2. (Cauchy - Peano Existence Theorem)

Let D be an open subset of $\mathbb{R} \times \mathbb{R}^n$, and $(t_0, x_0) \in D$.

Assume $f \in C(D, \mathbb{R}^n)$ and is locally Lipschitz in x on D .

Then there exists a $\gamma > 0$, s.t. IVP has a unique solution, with $|t - t_0| < \gamma$.

Maximal Interval of Existence

Step 0. $\exists x_0(t) = \varphi_0(t)$ on $[a_0, b_0]$.

Step 1. $x' = f(t, x)$. $x(a_0) = \varphi_0(a_0)$ on $[a_1, a_0]$. $\exists x = \varphi_-(t)$.

Step 2. $x' = f(t, x)$ $x(b_0) = \varphi_0(b_0)$ on $[b_0, b_1]$. $\exists x = \varphi_+(t)$.

Define $\varphi_1(t) = \begin{cases} \varphi_-(t), & t \in [a_1, a_0] \\ \varphi_0(t), & t \in [a_0, b_0] \\ \varphi_+(t), & t \in [b_0, b_1] \end{cases}$ called an extension of $\varphi_0(t)$ from $[a_0, b_0]$ to $[a_1, b_1]$

Let φ_{m+1} be an extension of φ_m from $[a_m, b_m]$ to $[a_{m+1}, b_{m+1}]$.

Let $\alpha = \lim_{m \rightarrow \infty} a_m$, $\beta = \lim_{m \rightarrow \infty} b_m$

Denote by $\varphi(t)$ the extension of $\varphi_0(t)$ on (α, β)

Lemma 1.4.1.

$$(a_m, \varphi(a_m)) \rightarrow (\alpha, x_\alpha).$$

(a) Assume $\{(a_m, \varphi(a_m))\}_{m=1}^{\infty}$ has an accumulation point 聚点.

(α, x_α) in D . Then $\lim_{t \rightarrow \alpha^+} \varphi(t) = x_\alpha$.

In this case, $\varphi(t)$ can be further extended to $t = \alpha$.

Pf. take $\varepsilon > 0$ small. $G_\varepsilon := \{ (t, x) : |t - \alpha| < \varepsilon, |x - x_\alpha| \leq \varepsilon \} \subset D$.

let $M_\varepsilon = \max \{ \max_{(t, x) \in G_\varepsilon} |f(t, x)|, 1 \}$.

then $\exists N \in \mathbb{N}$, s.t.

$$a_N - \alpha < \frac{\varepsilon}{2M_\varepsilon}, \quad |\varphi(a_N) - x_\alpha| < \frac{\varepsilon}{2}.$$

Claim that: (1.4.2.) $|\varphi(t) - \varphi(a_N)| \leq M_\varepsilon (a_N - \alpha)$, $\forall \alpha < t \leq a_N$

Assume the contrary, $\exists t^* \in (\alpha, a_N)$ s.t.

$$|\varphi(t) - \varphi(a_N)| < M_\varepsilon (a_N - \alpha) < \frac{\varepsilon}{2}, \quad \forall t^* < t \leq a_N$$

$$\text{and } |\varphi(t^*) - \varphi(a_N)| = M_\varepsilon (a_N - \alpha) \quad (*)$$

$$\text{Thus for } t^* \leq t \leq a_N, \quad |\varphi(t) - x_\alpha| \leq |\varphi(t) - \varphi(a_N)| + |\varphi(a_N) - x_\alpha| < \varepsilon.$$

$$\Rightarrow |\varphi(t^*) - \varphi(a_N)| = \left| \int_{t^*}^{a_N} f(s, \varphi) ds \right|$$

$$\leq M_\varepsilon (a_N - t^*)$$

$$< M_\varepsilon (a_N - \alpha).$$

which contradicts (*).

